Research Brief: Retrieval-Augmented Generation

Executive Summary

Retrieval-Augmented Generation (RAG) combines information retrieval with large language model generation to produce factual, grounded answers. This brief outlines the architecture decisions for our research agent platform.

Key Findings

1. Local embedding models (all-MiniLM-L6-v2) achieve 87% answer relevance on internal document benchmarks without API costs.
2. Format-aware chunking improves retrieval precision by 12% over fixed-size splitting.
3. Citation-enforced prompting reduces hallucination rate to under 5% in manual evaluation.
4. pgvector with IVFFlat indexing handles 100K+ chunks with sub-50ms retrieval latency.

Recommendations

- Default to local Ollama inference for development; use OpenAI for production QA.
- Set retrieval top-K to 5 with a 0.3 similarity threshold.
- Archive all uploaded documents for compliance audit trails.
- Evaluate with RAGAS metrics before production deployment.

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